

MATH0005 Algebra 1 — Practice Problems & Solutions (Set 2)

Problem 1 — Matrix Invertibility and Parametric Systems

Let

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 4 & \mu & 1 \\ -2 & 3 & \mu \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 5 \\ 1 \end{pmatrix}.$$

- (a) Find all values of $\mu \in \mathbb{R}$ for which A is not invertible by computing $\det A$.
- (b) For each value of μ found in (a), determine whether the system $A\mathbf{x} = \mathbf{b}$ has no solution or infinitely many solutions. For the case(s) with infinitely many solutions, find the general solution expressed as a particular solution plus the fundamental solutions of $A\mathbf{x} = \mathbf{0}$.
- (c) For $\mu = 3$ (a value not found in (a)), compute A^{-1} using the RREF method and hence solve $A\mathbf{x} = \mathbf{b}$.

Solution

Part (a): Finding singular values of μ Expanding $\det A$ along the first row:

$$\begin{aligned} \det A &= 2 \det \begin{pmatrix} \mu & 1 \\ 3 & \mu \end{pmatrix} - 1 \det \begin{pmatrix} 4 & 1 \\ -2 & \mu \end{pmatrix} + (-1) \det \begin{pmatrix} 4 & \mu \\ -2 & 3 \end{pmatrix} \\ &= 2(\mu^2 - 3) - (4\mu + 2) - (12 + 2\mu) \\ &= 2\mu^2 - 6 - 4\mu - 2 - 12 - 2\mu \\ &= 2\mu^2 - 6\mu - 20 = 2(\mu^2 - 3\mu - 10) = 2(\mu - 5)(\mu + 2). \end{aligned}$$

So A is **not invertible** when $\boxed{\mu = 5}$ or $\boxed{\mu = -2}$.

Part (b): Consistency analysis at singular values **Case $\mu = 5$:**

$$[A|\mathbf{b}] = \left(\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 4 & 5 & 1 & 5 \\ -2 & 3 & 5 & 1 \end{array} \right).$$

R2 $\rightarrow$ R2 $-$ 2R1, R3 $\rightarrow$ R3 $+$ R1:

$$\left(\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 0 & 3 & 3 & 3 \\ 0 & 4 & 4 & 2 \end{array} \right).$$

R2 $\rightarrow$ $\frac{1}{3}$ R2:

$$\left(\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 4 & 4 & 2 \end{array} \right).$$

R3 $\rightarrow$ R3 $-$ 4R2:

$$\left(\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & -2 \end{array} \right).$$

The last row reads $0 = -2$. **No solution** when $\mu = 5$.

Case $\mu = -2$:

$$[A|\mathbf{b}] = \left(\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 4 & -2 & 1 & 5 \\ -2 & 3 & -2 & 1 \end{array} \right).$$

R2 $\rightarrow$ R2 $-$ 2R1, R3 $\rightarrow$ R3 $+$ R1:

$$\left(\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 0 & -4 & 3 & 3 \\ 0 & 4 & -3 & 2 \end{array} \right).$$

R3 $\rightarrow$ R3 $+$ R2:

$$\left(\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 0 & -4 & 3 & 3 \\ 0 & 0 & 0 & 5 \end{array} \right).$$

Again the last row gives $0 = 5$. **No solution** when $\mu = -2$.

Remark: Both singular values yield inconsistent systems; neither gives infinitely many solutions. Part (b) is therefore vacuous for the general-solution request, but the full row reduction is required to establish this.

Part (c): Inverse at $\mu = 3$ With $\mu = 3$:

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 4 & 3 & 1 \\ -2 & 3 & 3 \end{pmatrix}.$$

Row reduce $[A \mid I_3]$:

$$\left(\begin{array}{ccc|ccc} 2 & 1 & -1 & 1 & 0 & 0 \\ 4 & 3 & 1 & 0 & 1 & 0 \\ -2 & 3 & 3 & 0 & 0 & 1 \end{array} \right).$$

$R2 \rightarrow R2 - 2R1, R3 \rightarrow R3 + R1$:

$$\left(\begin{array}{ccc|ccc} 2 & 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 3 & -2 & 1 & 0 \\ 0 & 4 & 2 & 1 & 0 & 1 \end{array} \right).$$

$R3 \rightarrow R3 - 4R2$:

$$\left(\begin{array}{ccc|ccc} 2 & 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 3 & -2 & 1 & 0 \\ 0 & 0 & -10 & 9 & -4 & 1 \end{array} \right).$$

$R3 \rightarrow -\frac{1}{10}R3$:

$$\left(\begin{array}{ccc|ccc} 2 & 1 & -1 & 1 & 0 & 0 \\ 0 & 1 & 3 & -2 & 1 & 0 \\ 0 & 0 & 1 & -9/10 & 2/5 & -1/10 \end{array} \right).$$

$R2 \rightarrow R2 - 3R3, R1 \rightarrow R1 + R3$:

$$\left(\begin{array}{ccc|ccc} 2 & 1 & 0 & 1/10 & 2/5 & -1/10 \\ 0 & 1 & 0 & 7/10 & -1/5 & 3/10 \\ 0 & 0 & 1 & -9/10 & 2/5 & -1/10 \end{array} \right).$$

$R1 \rightarrow R1 - R2$, then $R1 \rightarrow \frac{1}{2}R1$:

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -3/10 & 3/10 & -1/5 \\ 0 & 1 & 0 & 7/10 & -1/5 & 3/10 \\ 0 & 0 & 1 & -9/10 & 2/5 & -1/10 \end{array} \right).$$

Therefore:

$$A^{-1} = \frac{1}{10} \begin{pmatrix} -3 & 3 & -2 \\ 7 & -2 & 3 \\ -9 & 4 & -1 \end{pmatrix}.$$

The solution to $A\mathbf{x} = \mathbf{b}$ is:

$$\mathbf{x} = A^{-1}\mathbf{b} = \frac{1}{10} \begin{pmatrix} -3 + 15 - 2 \\ 7 - 10 + 3 \\ -9 + 20 - 1 \end{pmatrix} = \frac{1}{10} \begin{pmatrix} 10 \\ 0 \\ 10 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

Problem 2 — Subspaces, Bases, and Dimension

(a) Let $V = \{\mathbf{x} \in \mathbb{R}^4 : x_1 + x_2 - x_3 = 0 \text{ and } x_2 + 2x_4 = 0\}$. Show that V is a subspace of $\mathbb{R}^4$, find a basis for V , and state $\dim V$.

(b) Let $W = \text{span}\{\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3\}$ where

$$\mathbf{w}_1 = \begin{pmatrix} 1 \\ 2 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{w}_2 = \begin{pmatrix} 2 \\ 4 \\ 1 \\ 3 \end{pmatrix}, \quad \mathbf{w}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}.$$

Find a basis for W and compute $\dim W$. Is $\mathbf{w}_1$ in the span of the remaining two vectors?

(c) Prove that $V \cap W$ is a subspace of $\mathbb{R}^4$. Without computing $V \cap W$ explicitly, use the dimension formula $\dim(V + W) = \dim V + \dim W - \dim(V \cap W)$ and the fact that $V + W \leq \mathbb{R}^4$ to find the possible values of $\dim(V \cap W)$.

Solution

Part (a): The subspace V $V = N(B)$ where $B = \begin{pmatrix} 1 & 1 & -1 & 0 \\ 0 & 1 & 0 & 2 \end{pmatrix}$. Kernels of matrices are always subspaces, so $V \leq \mathbb{R}^4$.

Row reduce B (already in echelon form with pivots in columns 1 and 2). Free variables: x_3, x_4 .

From row 2: $x_2 = -2x_4$. From row 1: $x_1 = -x_2 + x_3 = 2x_4 + x_3$.

Setting $(x_3, x_4) = (1, 0)$: $\mathbf{f}_1 = (1, 0, 1, 0)^T$.

Setting $(x_3, x_4) = (0, 1)$: $\mathbf{f}_2 = (2, -2, 0, 1)^T$.

Basis for V : $\{(1, 0, 1, 0)^T, (2, -2, 0, 1)^T\}$, so $\dim V = 2$.

Part (b): The subspace W Form the matrix with $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ as columns and row reduce:

$$\begin{pmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \\ 0 & 1 & 1 \\ 1 & 3 & 1 \end{pmatrix} \xrightarrow{R2 \rightarrow R2 - 2R1, R4 \rightarrow R4 - R1} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{pmatrix} \xrightarrow{R2 \leftrightarrow R3} \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

$R4 \rightarrow R4 - R2$:

$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Pivot columns are 1 and 2, so $\{\mathbf{w}_1, \mathbf{w}_2\}$ is a **basis for W** and $\dim W = 2$.

Since $\mathbf{w}_3$ is not a pivot column, it lies in the span of $\mathbf{w}_1$ and $\mathbf{w}_2$: indeed $\mathbf{w}_3 = \mathbf{w}_2 - \mathbf{w}_1 - (1, 2, 0, 1)^T$... let us verify directly. We need $a\mathbf{w}_1 + b\mathbf{w}_2 = \mathbf{w}_3$, i.e. $(a + 2b, 2a + 4b, b, a + 3b) = (0, 0, 1, 1)$. From row 3: $b = 1$; row 1: $a = -2$; check row 2: $-4 + 4 = 0$, row 4: $-2 + 3 = 1$.

So $\mathbf{w}_3 = -2\mathbf{w}_1 + \mathbf{w}_2 \in \text{span}\{\mathbf{w}_1, \mathbf{w}_2\}$.

Is $\mathbf{w}_1 \in \text{span}\{\mathbf{w}_2, \mathbf{w}_3\}$? Since $\mathbf{w}_1$ is a pivot vector, it is **not** in the span of the non-pivot columns (only $\mathbf{w}_3$ is non-pivot). More directly: $a\mathbf{w}_2 + b\mathbf{w}_3 = \mathbf{w}_1$ gives $(2a, 4a, a + b, 3a + b) = (1, 2, 0, 1)$, so $a = 1/2$, $a + b = 0 \Rightarrow b = -1/2$, check row 4: $3/2 - 1/2 = 1$. So in fact $\mathbf{w}_1 \in \text{span}\{\mathbf{w}_2, \mathbf{w}_3\}$. (This is consistent: $\{\mathbf{w}_2, \mathbf{w}_3\}$ is also a basis of W , as any two of the three generators span W since each is a linear combination of the others.)

Part (c): Dimension of $V \cap W$ $V \cap W$ is a subspace: For $\mathbf{u}, \mathbf{v} \in V \cap W$ and scalars α, β : since $\mathbf{u}, \mathbf{v} \in V$ and V is a subspace, $\alpha\mathbf{u} + \beta\mathbf{v} \in V$; similarly $\alpha\mathbf{u} + \beta\mathbf{v} \in W$. So $V \cap W$ is closed under linear combinations and contains $\mathbf{0}$, hence is a subspace. $\square$

Dimension bounds: $V + W \leq \mathbb{R}^4$ so $\dim(V + W) \leq 4$. Also $\dim(V + W) \geq \max(\dim V, \dim W) = 2$. The formula gives:

$$\dim(V \cap W) = \dim V + \dim W - \dim(V + W) = 2 + 2 - \dim(V + W).$$

Since $2 \leq \dim(V + W) \leq 4$, we get $\dim(V \cap W) \in \{0, 1, 2\}$. The upper bound $\dim(V \cap W) \leq \min(\dim V, \dim W) = 2$ is also immediate from the subspace dimension theorem.

Problem 3 — Linear Maps, Rank-Nullity, and the Change of Basis Theorem

Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear map given by

$$T \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x + y + z \\ x - y + 2z \\ 2x + 3z \end{pmatrix}.$$

(a) Write down the matrix A such that $T = T_A$ (multiplication by A with respect to the standard basis). Find the rank and nullity of T , and find a basis for $\ker T$ and $\operatorname{Im} T$.

(b) Let $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_3\}$ be the basis

$$\mathbf{b}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{b}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{b}_3 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

Find the change-of-basis matrix P for $\mathcal{B}$, and use the change-of-basis theorem to compute the matrix $P^{-1}AP$ representing T with respect to $\mathcal{B}$ on both domain and codomain.

(c) Use the formula $(P^{-1}AP)^n = P^{-1}A^nP$ to find a simple expression for A^n in the case that A is diagonalizable. (You do not need to check diagonalizability; simply carry out the computation using your answer to (b), or explain why it is or is not useful here.)

Solution

Part (a): Matrix, rank, nullity Reading off columns:

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ 2 & 0 & 3 \end{pmatrix}.$$

Row reduce:

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & -2 & 1 \\ 0 & -2 & 1 \end{pmatrix} \xrightarrow{R3 \rightarrow R3 - R2} \begin{pmatrix} 1 & 1 & 1 \\ 0 & -2 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Two pivots, one free variable (z). **Rank** = 2, **Nullity** = 1.

Kernel: From R2: $y = z/2$. From R1: $x = -y - z = -3z/2$. Setting $z = 2$: $\ker T = \text{span}\{(-3, 1, 2)^T\}$.

Image: Pivot columns 1 and 2 of A form a basis:

$$\text{Im } T = \text{span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} \right\}.$$

Part (b): Change-of-basis matrix

$$P = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

Find P^{-1} using RREF (or directly since $\det P = 1(1-0) - 0 + 1(1-0) = 2$):

Row reduce $[P \mid I]$:

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right) \xrightarrow{R2 \rightarrow R2 - R1} \left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right).$$

$R3 \rightarrow R3 - R2$:

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 1 & 0 \\ 0 & 0 & 2 & 1 & -1 & 1 \end{array} \right).$$

$R3 \rightarrow \frac{1}{2}R3$; $R1 \rightarrow R1 - R3$; $R2 \rightarrow R2 + R3$:

$$P^{-1} = \frac{1}{2} \begin{pmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix}.$$

Now compute $P^{-1}AP$ by first computing AP :

$$AP = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ 2 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 & 2 \\ 0 & 1 & 3 \\ 2 & 3 & 5 \end{pmatrix}.$$

$$\text{Then } P^{-1}(AP) = \frac{1}{2} \begin{pmatrix} 1 & 1 & -1 \\ -1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 2 & 2 \\ 0 & 1 & 3 \\ 2 & 3 & 5 \end{pmatrix}:$$

Row 1: $\frac{1}{2}(2 + 0 - 2, 2 + 1 - 3, 2 + 3 - 5) = \frac{1}{2}(0, 0, 0) = (0, 0, 0)$.

Row 2: $\frac{1}{2}(-2 + 0 + 2, -2 + 1 + 3, -2 + 3 + 5) = \frac{1}{2}(0, 2, 6) = (0, 1, 3)$.

Row 3: $\frac{1}{2}(2 + 0 + 2, 2 - 1 + 3, 2 - 3 + 5) = \frac{1}{2}(4, 4, 4) = (2, 2, 2)$.

$$P^{-1}AP = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 3 \\ 2 & 2 & 2 \end{pmatrix}.$$

Part (c): Powers of A The matrix $P^{-1}AP$ above is not diagonal, so we cannot directly use the diagonalization formula $A^n = PD^nP^{-1}$ with this particular basis. The matrix A has rank 2 and nullity 1, so 0 is an eigenvalue. To apply the diagonalization formula, one would instead find the full eigendecomposition of A (a separate computation yielding diagonal D), and then write $A^n = PD^nP^{-1}$ where the columns of P are eigenvectors. The identity $(P^{-1}AP)^n = P^{-1}A^nP$ holds for **any** invertible P ; it is particularly useful when $P^{-1}AP = D$ is diagonal, since $D^n = \text{diag}(\lambda_1^n, \dots, \lambda_k^n)$ is trivial to compute.

Problem 4 — Determinants, Eigenvalues, and Diagonalization over $\mathbb{C}$

Let

$$B = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

(a) Compute the characteristic polynomial $c_B(t) = \det(tI - B)$ and find all eigenvalues of B over $\mathbb{C}$.

(b) For each eigenvalue, find a basis for the eigenspace $E_\lambda(B) \subseteq \mathbb{C}^3$.

(c) Determine whether B is diagonalizable over $\mathbb{R}$. Is it diagonalizable over $\mathbb{C}$? If so, find invertible P and diagonal D with entries in $\mathbb{C}$ such that $P^{-1}BP = D$.

(d) Using your diagonalization, compute B^{2026} .

Solution

Part (a): Characteristic polynomial

$$c_B(t) = \det \begin{pmatrix} t & 1 & 0 \\ -1 & t & 0 \\ 0 & 0 & t-2 \end{pmatrix} = (t-2) \det \begin{pmatrix} t & 1 \\ -1 & t \end{pmatrix} = (t-2)(t^2 + 1).$$

Eigenvalues over $\mathbb{C}$: $\lambda_1 = 2, \lambda_2 = i, \lambda_3 = -i$.

Since $t^2 + 1$ has no real roots, B has only one real eigenvalue ($\lambda = 2$).

Part (b): Eigenspaces $\lambda = 2$: $B - 2I = \begin{pmatrix} -2 & -1 & 0 \\ 1 & -2 & 0 \\ 0 & 0 & 0 \end{pmatrix}$.

Row reduce: $R1 \leftrightarrow R2$: $\begin{pmatrix} 1 & -2 & 0 \\ -2 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. $R2 \rightarrow R2 + 2R1$: $\begin{pmatrix} 1 & -2 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. So

$y = 0, x = 0, z$ free.

Basis: $\{(0, 0, 1)^T\}$. (Equivalently $\mathbf{e}_3$.)

$\lambda = i$: $B - iI = \begin{pmatrix} -i & -1 & 0 \\ 1 & -i & 0 \\ 0 & 0 & 2 - i \end{pmatrix}$.

Note $R1 = i \cdot R2$ (check: $i \cdot 1 = i \neq -i$; rather $R1 = (-i) \cdot R2$? $(-i)(1) = -i$, $(-i)(-i) = i^2 = -1$). So rows 1 and 2 are dependent. Also $2 - i \neq 0$ so $z = 0$.

From row 2: $x - iy = 0 \Rightarrow x = iy$.

Basis: $\{(i, 1, 0)^T\}$.

$\lambda = -i$: By conjugation (or direct computation): basis $\{(-i, 1, 0)^T\}$.

Part (c): Diagonalizability Over $\mathbb{R}$: Only one real eigenvalue ($\lambda = 2$) with a 1-dimensional eigenspace. Since $\sum \dim E_\lambda = 1 < 3$, B is **not diagonalizable over $\mathbb{R}$** .

Over $\mathbb{C}$: Three distinct eigenvalues $2, i, -i$, so three linearly independent eigenvectors. B is **diagonalizable over $\mathbb{C}$** .

$$P = \begin{pmatrix} 0 & i & -i \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & i & 0 \\ 0 & 0 & -i \end{pmatrix}.$$

Part (d): B^{2026} Since $P^{-1}BP = D$, we have $B^n = PD^nP^{-1}$. Now $D^{2026} = \text{diag}(2^{2026}, i^{2026}, (-i)^{2026})$.

$$i^{2026} = i^{4 \cdot 506 + 2} = (i^4)^{506} \cdot i^2 = 1 \cdot (-1) = -1.$$

$$\text{Similarly } (-i)^{2026} = ((-1)^{2026})(i^{2026}) = 1 \cdot (-1) = -1.$$

$$\text{So } D^{2026} = \text{diag}(2^{2026}, -1, -1).$$

To find $B^{2026} = PD^{2026}P^{-1}$, observe that B acts as a **rotation by 90°** in the xy -plane and as multiplication by 2 in the z -direction. More explicitly, for the top-left 2×2 block:

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^{2026} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^{4 \cdot 506 + 2} = \left[\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^4 \right]^{506} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}^2 = I^{506} \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I_2.$$

Therefore:

$$B^{2026} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2^{2026} \end{pmatrix}.$$

Problem 5 — Orthogonal Matrices, Gram–Schmidt, and the Spectral Theorem

(a) Let

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \\ -1 \end{pmatrix}.$$

Apply Gram–Schmidt to $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ to obtain an orthonormal set $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$.

(b) Let C be the real symmetric matrix

$$C = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 3 \end{pmatrix}.$$

Find the characteristic polynomial of C , determine all eigenvalues, and find an orthonormal basis of each eigenspace.

(c) Hence write down an orthogonal matrix Q and diagonal matrix Λ such that $Q^T C Q = \Lambda$. Verify that Q is orthogonal by checking $Q Q^T = I$.

(d) Using the spectral decomposition $C = Q \Lambda Q^T$, express C as a sum of rank-one matrices (outer products of eigenvectors), and use this to compute C^{-1} .

Solution

Part (a): Gram–Schmidt Step 1. $\mathbf{u}_1 = \mathbf{v}_1 = (1, 0, 1, 0)^T$, $\|\mathbf{u}_1\|^2 = 2$.

Step 2. $\mathbf{v}_2 \cdot \mathbf{u}_1 = 1 + 0 + 0 + 0 = 1$.

$$\mathbf{u}_2 = \mathbf{v}_2 - \frac{1}{2}\mathbf{u}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 1/2 \\ 0 \\ 1/2 \\ 0 \end{pmatrix} = \begin{pmatrix} 1/2 \\ 1 \\ -1/2 \\ 1 \end{pmatrix}.$$

$$\|\mathbf{u}_2\|^2 = 1/4 + 1 + 1/4 + 1 = 5/2.$$

Step 3. $\mathbf{v}_3 \cdot \mathbf{u}_1 = 0 + 0 + 1 + 0 = 1$. $\mathbf{v}_3 \cdot \mathbf{u}_2 = 0 + 1 - 1/2 - 1 = -1/2$.

$$\mathbf{u}_3 = \mathbf{v}_3 - \frac{1}{2}\mathbf{u}_1 - \frac{-1/2}{5/2}\mathbf{u}_2 = \mathbf{v}_3 - \frac{1}{2}\mathbf{u}_1 + \frac{1}{5}\mathbf{u}_2.$$

$$= \begin{pmatrix} 0 \\ 1 \\ 1 \\ -1 \end{pmatrix} - \begin{pmatrix} 1/2 \\ 0 \\ 1/2 \\ 0 \end{pmatrix} + \frac{1}{5} \begin{pmatrix} 1/2 \\ 1 \\ -1/2 \\ 1 \end{pmatrix} = \begin{pmatrix} -1/2 \\ 1 \\ 1/2 \\ -1 \end{pmatrix} + \begin{pmatrix} 1/10 \\ 1/5 \\ -1/10 \\ 1/5 \end{pmatrix} = \begin{pmatrix} -2/5 \\ 6/5 \\ 2/5 \\ -4/5 \end{pmatrix}.$$

Scale by 5: direction $(-2, 6, 2, -4)^T = 2(-1, 3, 1, -2)^T$. So $\mathbf{u}_3 = (-1, 3, 1, -2)^T$ (up to scaling). $\|\mathbf{u}_3'\|^2 = 1 + 9 + 1 + 4 = 15$ where $\mathbf{u}_3' = (-1, 3, 1, -2)^T$.

Normalising:

$$\mathbf{e}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 \\ 2 \\ -1 \\ 2 \end{pmatrix}, \quad \mathbf{e}_3 = \frac{1}{\sqrt{15}} \begin{pmatrix} -1 \\ 3 \\ 1 \\ -2 \end{pmatrix}.$$

(Check: $\mathbf{u}_2$ scaled by 2 gives $(1, 2, -1, 2)^T$ with $\|(1, 2, -1, 2)\|^2 = 1 + 4 + 1 + 4 = 10$.)

Part (b): Eigenvalues of C

$$c_C(t) = \det(tI - C) = \det \begin{pmatrix} t-2 & 0 & -1 \\ 0 & t-2 & -1 \\ -1 & -1 & t-3 \end{pmatrix}.$$

Expand along column 1:

$$\begin{aligned} &= (t-2) \det \begin{pmatrix} t-2 & -1 \\ -1 & t-3 \end{pmatrix} - (-1) \det \begin{pmatrix} 0 & -1 \\ -1 & t-3 \end{pmatrix} \\ &= (t-2)[(t-2)(t-3) - 1] + [0 \cdot (t-3) - 1] \end{aligned}$$

$$= (t-2)(t^2 - 5t + 5) - 1.$$

Expanding: $(t-2)(t^2 - 5t + 5) = t^3 - 5t^2 + 5t - 2t^2 + 10t - 10 = t^3 - 7t^2 + 15t - 10$.

$$c_C(t) = t^3 - 7t^2 + 15t - 10 - 1 = t^3 - 7t^2 + 15t - 11.$$

Test $t = 1$: $1 - 7 + 15 - 11 = -2 \neq 0$. Test $t = 11$: too large. Try rational roots.

Test $t = 1$ again: no. Test $t = 11/1$, nope.

Let us recompute more carefully. Expanding along the first row instead:

$$c_C(t) = (t-2) \det \begin{pmatrix} t-2 & -1 \\ -1 & t-3 \end{pmatrix} - 0 + (-1) \det \begin{pmatrix} 0 & t-2 \\ -1 & -1 \end{pmatrix}.$$

$$= (t-2)[(t-2)(t-3) - 1] - 1 \cdot [0 \cdot (-1) - (t-2)(-1)]$$

$$= (t-2)(t^2 - 5t + 5) - (t-2) = (t-2)(t^2 - 5t + 5 - 1) = (t-2)(t^2 - 5t + 4).$$

$$= (t-2)(t-1)(t-4).$$

Eigenvalues: $\lambda_1 = 1$, $\lambda_2 = 2$, $\lambda_3 = 4$. (All distinct, so C is diagonalizable over $\mathbb{R}$, as expected by the Spectral Theorem since C is symmetric.)

Eigenspace $\lambda = 1$: Row reduce $C - I$:

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix} \xrightarrow{R3 \rightarrow R3 - R1 - R2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

x_3 free; $x_1 = -x_3$, $x_2 = -x_3$. Basis: $\{(-1, -1, 1)^T\}$. Normalised: $\frac{1}{\sqrt{3}}(-1, -1, 1)^T$.

Eigenspace $\lambda = 2$: Row reduce $C - 2I$:

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R1 \rightarrow R1 - R2} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

x_2 free, $x_3 = 0$, $x_1 = -x_2$. Basis: $\{(-1, 1, 0)^T\}$. Normalised: $\frac{1}{\sqrt{2}}(-1, 1, 0)^T$.

Eigenspace $\lambda = 4$: Row reduce $C - 4I$:

$$\begin{pmatrix} -2 & 0 & 1 \\ 0 & -2 & 1 \\ 1 & 1 & -1 \end{pmatrix} \xrightarrow{R1 \leftrightarrow R3} \begin{pmatrix} 1 & 1 & -1 \\ 0 & -2 & 1 \\ -2 & 0 & 1 \end{pmatrix} \xrightarrow{R3 \rightarrow R3+2R1} \begin{pmatrix} 1 & 1 & -1 \\ 0 & -2 & 1 \\ 0 & 2 & -1 \end{pmatrix} \xrightarrow{R3 \rightarrow R3+R2} \begin{pmatrix} 1 & 1 & -1 \\ 0 & -2 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

From R2: $x_2 = x_3/2$. From R1: $x_1 = -x_2 + x_3 = x_3/2$. Setting $x_3 = 2$: $(1, 1, 2)^T$. Normalised: $\frac{1}{\sqrt{6}}(1, 1, 2)^T$.

Part (c): Orthogonal diagonalization

$$Q = \begin{pmatrix} \frac{-1}{\sqrt{3}} & \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{-1}{\sqrt{3}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & 0 & \frac{2}{\sqrt{6}} \end{pmatrix}, \quad \Lambda = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{pmatrix}.$$

Verification $QQ^T = I$: The columns of Q are $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3$ (eigenvectors, normalised). By the Spectral Theorem, eigenvectors of a real symmetric matrix for distinct eigenvalues are orthogonal (proved in part (c) of Problem 4 in the previous set). Direct check: $\mathbf{q}_1 \cdot \mathbf{q}_2 = \frac{1}{\sqrt{6}}(1-1+0) = 0$; $\mathbf{q}_1 \cdot \mathbf{q}_3 = \frac{1}{\sqrt{18}}(-1-1+2) = 0$; $\mathbf{q}_2 \cdot \mathbf{q}_3 = \frac{1}{\sqrt{12}}(-1+1+0) = 0$. Each column has unit norm by construction. Hence $QQ^T = I$. $\square$

Part (d): Spectral decomposition and C^{-1} The spectral decomposition is:

$$C = \lambda_1 \mathbf{q}_1 \mathbf{q}_1^T + \lambda_2 \mathbf{q}_2 \mathbf{q}_2^T + \lambda_3 \mathbf{q}_3 \mathbf{q}_3^T = 1 \cdot \mathbf{q}_1 \mathbf{q}_1^T + 2 \cdot \mathbf{q}_2 \mathbf{q}_2^T + 4 \cdot \mathbf{q}_3 \mathbf{q}_3^T.$$

Each $\mathbf{q}_i \mathbf{q}_i^T$ is a rank-one projection matrix. Since C is symmetric positive definite (all eigenvalues > 0):

$$C^{-1} = Q\Lambda^{-1}Q^T = \frac{1}{\lambda_1} \mathbf{q}_1 \mathbf{q}_1^T + \frac{1}{\lambda_2} \mathbf{q}_2 \mathbf{q}_2^T + \frac{1}{\lambda_3} \mathbf{q}_3 \mathbf{q}_3^T = \mathbf{q}_1 \mathbf{q}_1^T + \frac{1}{2} \mathbf{q}_2 \mathbf{q}_2^T + \frac{1}{4} \mathbf{q}_3 \mathbf{q}_3^T.$$

Computing each outer product:

$$\mathbf{q}_1 \mathbf{q}_1^T = \frac{1}{3} \begin{pmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix}, \quad \mathbf{q}_2 \mathbf{q}_2^T = \frac{1}{2} \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{q}_3 \mathbf{q}_3^T = \frac{1}{6} \begin{pmatrix} 1 & 1 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 4 \end{pmatrix}.$$

$$C^{-1} = \frac{1}{3} \begin{pmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix} + \frac{1}{4} \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \frac{1}{24} \begin{pmatrix} 1 & 1 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 4 \end{pmatrix}.$$

Combining (common denominator 24):

$$C^{-1} = \frac{1}{24} \left[\begin{pmatrix} 8 & 8 & -8 \\ 8 & 8 & -8 \\ -8 & -8 & 8 \end{pmatrix} + \begin{pmatrix} 6 & -6 & 0 \\ -6 & 6 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 1 & 2 \\ 1 & 1 & 2 \\ 2 & 2 & 4 \end{pmatrix} \right] = \frac{1}{24} \begin{pmatrix} 15 & 3 & -6 \\ 3 & 15 & -6 \\ -6 & -6 & 12 \end{pmatrix} = \frac{1}{8} \begin{pmatrix} 5 & 1 & -2 \\ 1 & 5 & -2 \\ -2 & -2 & 4 \end{pmatrix}$$

$$\mathbf{Check:} \ C \cdot C^{-1} = \frac{1}{8} \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 3 \end{pmatrix} \begin{pmatrix} 5 & 1 & -2 \\ 1 & 5 & -2 \\ -2 & -2 & 4 \end{pmatrix} = \frac{1}{8} \begin{pmatrix} 10+0-2 & 2+0-2 & -4+0+4 \\ 0+2-2 & 0+10-2 & 0-4+4 \\ 5+1-6 & 1+5-6 & -2-2+12 \end{pmatrix} =$$

$$\frac{1}{8} \begin{pmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 8 \end{pmatrix} = I.$$